

continuation

Continuation Document

Revision: 41 **Last modified:** 2026-06-01T13:04:06Z **Description:** Sacred invariant resumption document for Helix Cluster OS **Authority:** Constitution §12.10
Maintainer: Operator + AI loop

§0: How to use this document

Any CLI agent resuming work on this project MUST read this file first.

§1: Recently Completed Work (last 10 commits)

Commit	Message
44db918	Foundation wave 4 + cross-platform parity: 9 real-infra items + macOS GPU/resources + real VMSSH — integration-proven
366a436	CLAUDE-2: Cross-Platform Parity Guarantee — no Linux-only feature may fall back to a mock/stub on macOS
f6209f6	Foundation wave 3b: mark 7 items Completed (HXC-1010/1011/1012/1013/1015/1020/1028)
85ee85e	Foundation wave 3b: real-infra impl (infra orchestrator / podman builder / vault api) — integration-proven
909d107	Disable all active CI per operator directive: park race.yml in disabled/
7cc8147	Foundation wave 3a: mark 6 items Completed (HXC-1019/1022/1023/1025/1029/1038)
32feec3	Foundation wave 3a: real fixes for lock leak + build sync/cancel + security/wasm coverage
930a7ac	Foundation wave 2: mark 30 items Completed (HXC-1039..1041/1046..1061/1064..1074)
a084eb1	Foundation wave 2: anti-bluff harden 29 pkg test items + §1.1 mutation runner + -race CI gate
3fb815d	Foundation wave 1: mark 30 verified items Completed (HXC-1001..1009/1016-1018/1021/1024/1026-1027/1030-1037/1042-1045/1062-1063)

§2: Environment Snapshot

Property	Value
Branch	main
Commit	44db918
Timestamp	2026-06-01T13:04:06Z

§3: Active Work

HXC	Title	Status
HXC-922	Node service, advisory locks, chaos tests, docs	✓ Done
HXC-921	Web UI, K8s/Helm, integration, E2E, benchmarks	✓ Done
HXC-920	Health, security, build, wireguard services	✓ Done
HXC-919	htmx, messaging, GPU, WASM, storage, build manifest	✓ Done
HXC-918	Wire scheduler/session gRPC to real backends, etcd discovery	✓ Done
HXC-916	Stub expansion (config, retry, ratelimit, validator, build cache)	✓ Done
HXC-914	LLM, policy, setup, distributed lock	✓ Done
HXC-913	Health, security, ClassAds parser	✓ Done
HXC-912	Gateway, helixd, helix-agent	✓ Done
HXC-911	Scheduler, Session gRPC services	✓ Done
HXC-910	Console, security, testing infra	✓ Done

§4: Next Planned Work

All MVP, Phase 2, Phase 3, and Phase 4 are **COMPLETE**. Potential future enhancements:

1. **Performance optimization** — Profile and optimize hot paths
2. **Multi-region support** — Cross-region cluster federation
3. **GPU scheduling v2** — Multi-GPU, fractional GPU allocation

4. **Web UI real-time** — WebSocket integration for live updates
5. **Operator pattern** — Kubernetes operator for cluster management

§5: Known Issues / Blockers

- None

§6: Quick Commands

Run all tests (requires GOWORK due to workspace modules)

```
GOWORK=$(pwd)/go.work go test ./... -race -count=1
```

Build all binaries

```
GOWORK=$(pwd)/go.work go build ./cmd/...
```

Build web UI

```
cd web && npm run build
```

Render Helm chart

```
helm template helix-cluster deploy/helm/
```

Run benchmarks

```
go test -bench=. ./test/benchmark
```

Run chaos tests

```
go test -race -count=1 ./test/chaos
```